MAT 319 HW01

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Problem (1.5)**.** Prove $1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^n} = 2 - \frac{1}{2^n}$ for all positive integers n.

Solution. We show this by induction on n. For the base case of $n = 1$, we get the obviously true equation

$$\frac{3}{2} = 1 + \frac{1}{2^1} = 2 - \frac{1}{2^1} = \frac{3}{2}.$$

Assume as the inductive hypothesis that

$$1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^n} = 2 - \frac{1}{2^n}$$

for some $n \in \mathbb{N}$. Then, we want to show that the equation holds when n is replaced with $n + 1$, i.e.

$$1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^{n+1}} = 2 - \frac{1}{2^{n+1}}.$$

This can be done by manipulating the left-hand side as follows:

$$\begin{aligned}\left(1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^n}\right) + \frac{1}{2^{n+1}} &= 2 - \frac{1}{2^n} + \frac{1}{2^{n+1}} \\ &= 2 - \frac{2}{2^{n+1}} + \frac{1}{2^{n+1}} \\ &= 2 - \frac{1}{2^{n+1}}.\end{aligned}$$

Therefore, by the principle of mathematical induction,

$$1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^n} = 2 - \frac{1}{2^n}$$

for all positive integers n. □

Problem (2.3)**.** Show $\sqrt{2 + \sqrt{2}}$ is not a rational number.

We present two solutions.

Solution 1. Suppose for the sake of contradiction $\sqrt{2 + \sqrt{2}}$ is rational. Then, it can be expressed as a ratio of integers $\frac{a}{b}$ with $b \neq 0$. By squaring and subtracting two, we find that

$$\sqrt{2} = \frac{a^2}{b^2} - 2 = \frac{a^2 - 2b^2}{b^2}$$

which is rational because a and b being integers implies that $a^2 - 2b^2$ and b^2 are both integers. However, this is a contradiction because $\sqrt{2}$ is irrational, so $\sqrt{2 + \sqrt{2}}$ must be irrational. □

Solution 2. Observe that $\sqrt{2+\sqrt{2}}$ is a solution to the polynomial equation

$$(x^2-2)^2-2=0.$$

Expanding the left-hand side gives $x^4-4x^2+4-2=x^4-4x^2+2=0$. By the rational root theorem (Corollary 2.3 in the textbook), any rational value of x satisfying the equation must be an integer dividing the constant term -2. Thus, if $\sqrt{2+\sqrt{2}}$ is rational, then, being a root of x^4-4x^2+2, it must be an integer. However, $1<\sqrt{2+\sqrt{2}}<2$ because

$$1<2+\sqrt{2}<4 \iff -1<\sqrt{2}<2,$$

and the latter inequality is true, so $\sqrt{2+\sqrt{2}}$ is bounded between two consecutive integers and thus not an integer, hence it is irrational. $\square$

Problem (3.4)**.** Prove (v) and (vii) of Theorem 3.2.

Solution. We can prove (v) just by taking $a=1$ in property (iv), as $1=1^2\geq 0$. We do have to check that $1^2\neq 0$; this follows from property (vi) of Theorem 3.1, as $1\cdot 1=0$ would imply $1=0$, but obviously this is not true.

To prove (vii), we notice that a and b being positive means $\frac{1}{a}$, $\frac{1}{b}$, and $\frac{1}{ab}$ are all positive too due to properties (iii) and (vi) in 3.2. This means that

$$0\cdot\frac{1}{ab}<a\cdot\frac{1}{ab}<b\cdot\frac{1}{ab} \iff 0<\frac{1}{a}<\frac{1}{b},$$

as desired. $\square$

Problem (3.6)**.**

(a) Prove $|a+b+c|\leq|a|+|b|+|c|$ for all a, b, $c\in\mathbb{R}$.

(b) Use induction to prove

$$|a_1+a_2+\cdots+a_n|\leq|a_1|+|a_2|+\cdots+|a_n|$$

for n real numbers a_1, a_2, $\ldots$, a_n.

Solution. For part (a), apply triangle inequality twice:

$$(|a|+|b|)+|c|\geq|a+b|+|c|\geq|a+b+c|.$$

For part (b), we induct on n, as instructed, with the base case of $n=1$ being the trivial statement $|a_1|\leq|a_1|$.

Now, assume as the inductive hypothesis that

$$|a_1+a_2+\cdots+a_n|\leq|a_1|+|a_2|+\cdots+|a_n|.$$

We wish to prove that

$$|a_1+a_2+\cdots+a_{n+1}|\leq|a_1|+|a_2|+\cdots+|a_{n+1}|,$$

which is done exactly as in part (a):

$$|a_1|+|a_2|+\cdots+|a_n|+|a_{n+1}|\geq|a_1+a_2+\cdots+a_n|+|a_{n+1}|\geq|a_1+a_2+\cdots+a_{n+1}|,$$

with the first inequality following from the hypothesis and the second from the two-variable version of the inequality.

We have hence proven that $|a_1+\cdots+a_n|\leq|a_1|+\cdots+|a_n|$ for all positive integers n using the principle of mathematical induction. $\square$

> **Remark.** The triangle inequality can also be proven non-inductively by squaring both sides and subtracting $a_1^2 + a_2^2 + \cdots + a_n^2$ to get
>
> $$|a_1 + a_2 + \cdots + a_n| \le |a_1| + |a_2| + \cdots + |a_n| \iff \sum_{1 \le i < j \le n} 2a_i a_j \le \sum_{1 \le i < j \le n} 2|a_i a_j|$$
>
> which is obviously true because $|x| \ge x$ for all real x.

Problem (4.11)**.** Consider a, $b \in \mathbb{R}$ where $a < b$. Use the denseness of $\mathbb{Q}$ in $\mathbb{R}$ to show there are infinitely many rationals between a and b.

Solution. The main idea is to split (a, b) into infinitely many disjoint open intervals, each of which contains a rational number by denseness. The rest of this solution can be seen as a verification of the fact that this splitting is possible.

Define the sequence of real numbers (r_k) by $r_0 = a$ and

$$r_{k+1} = \frac{r_k + b}{2}$$

for integers $k \ge 0$.

> **Claim —** For all k, $r_k < r_{k+1}$. Additionally, the set $S = \{r_0, r_1, r_2, r_3, \dots\}$ is bounded below by a and bounded above by b.

Proof. We prove by induction on k that $r_k < r_{k+1} < b$ for all nonnegative integers k. For the base case of $k = 0$, we already know that $r_0 = a < b$, and then $r_0 < r_1 < b$ translates to

$$a < \frac{a+b}{2} < b \iff 2a = (a + a) < a + b < b + b$$

We then want to show that if $r_k < r_{k+1} < b$, then $r_{k+1} < r_{k+2} < b$. Using the definition of r_k, the inequality that we want to prove is

$$\frac{r_k + b}{2} < \frac{r_{k+1} + b}{2} < b,$$

which is equivalent to the inductive hypothesis after multiplying by 2 and subtracting b. Thus, by the principle of mathematical induction, $r_k < r_{k+1} < b$ for all k. This proves that b is an upper bound for S and that (r_k) is an increasing sequence.

Finally, every element of S is at least as large as $r_0 = a$ due to the previous paragraph, so $a = \min S$ is a lower bound for S. $\square$

Now, we are done, since, by density, there is a rational number in $(r_k, r_{k+1}) \subset (a, b)$ for every nonnegative integer k and the (r_k, r_{k+1}) intervals are pairwise disjoint (since the sequence is increasing) which gives us infinitely many rational numbers in the interval (a, b) as desired. $\square$

Problem (4.12)**.** Let $\mathbb{I}$ be the set of irrational numbers. Prove that if $a < b$, then there exists $x \in \mathbb{I}$ such that $a < x < b$.

Solution. Let d be a rational number satisfying $0 < d < b - a$, which must exist by denseness. By the archimedean property, we can find an integer n such that $a - \sqrt{2} < nd$. There must be a smallest such integer n because the right-hand side is not bounded below as n decreases;

let n_0 be this smallest value (in particular the inequality fails when $n \leq \frac{a-\sqrt{2}}{d}$, so n_0 is the smallest integer bigger than $\frac{a-\sqrt{2}}{2}$). Then, we claim that $n_0 d < b - \sqrt{2}$. We prove this by contradiction: if the inequality were false, then we find that

$$(n_0 - 1)d < a - \sqrt{2} < b - \sqrt{2} \leq n_0 d,$$

which implies the false statement

$$d = n_0 d - (n_0 - 1)d > (b - \sqrt{2}) - (a - \sqrt{2}) > b - a.$$

Therefore, we have found an integer n_0 such that

$$a < \sqrt{2} + n_0 d < b.$$

Since $\sqrt{2}$ is irrational, so is $\sqrt{2} + n_0 d$ (because $n_0 d$ is rational), so we have found an irrational number between a and b. □

Problem (5.1)**.** Write the following sets in interval notation:

(a) $\{x \in \mathbb{R} : x < 0\}$

(b) $\{x \in \mathbb{R} : x^3 \leq 8\}$

(c) $\{x^2 : x \in \mathbb{R}\}$

(d) $\{x \in \mathbb{R} : x^2 < 8\}$

Solution. **(a)** is the negative reals $(-\infty, 0)$.

(b) is $(-\infty, 2]$ because the inequality is equivalent to $x \leq 2$.

(c) is the nonnegative reals, since the square of a real number is always at least zero and all nonnegative real numbers are squares, so the corresponding interval is $[0, \infty)$.

(d) is $(-\sqrt{8}, \sqrt{8})$, since the inequality is equivalent to $|x| < \sqrt{8}$. □

Problem (5.2)**.** Give the infimum and supremum of each set listed in Exercise 5.1.

Solution. Reading off the intervals:

(a) infimum is $-\infty$, supremum is 0

(b) infimum is $-\infty$, supremum is 2

(c) infimum is 0, supremum is ∞

(d) infimum is $-\sqrt{8}$, supremum is $\sqrt{8}$

□